Decrease in Histidine-Rich Glycoprotein as a Novel Biomarker to Predict Sepsis Among Systemic Inflammatory Response Syndrome

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Objectives: Many biomarkers for sepsis are used in clinical practice; however, few have become the standard. We measured plasma histidine-rich glycoprotein levels in patients with systemic inflammatory response syndrome. We compared histidine-rich glycoprotein, procalcitonin, and presepsin levels to assess their significance as biomarkers.

Design: Single-center, prospective, observational cohort study.

Setting: ICU at an university-affiliated hospital.

Patients: Seventy-nine ICU patients (70 with systemic inflammatory response syndrome and 9 without systemic inflammatory response syndrome) and 16 healthy volunteers.

Interventions: None.

Measurements and Main Results: We collected blood samples from patients within 24 hours of ICU admission. Histidine-rich glycoprotein levels were determined using enzyme-linked immunosorbent assay. The median histidine-rich glycoprotein level in healthy volunteers ($n = 16$) was 63.00 µg/mL (interquartile range, 51.53–66.21 µg/mL). Histidine-rich glycoprotein levels in systemic inflammatory response syndrome patients ($n = 70$; 28.72 µg/mL [15.74–41.46 µg/mL]) were lower than those in nonsystemic inflammatory response syndrome patients ($n = 9$; 38.64 µg/mL [30.26–51.81 µg/mL]; $p = 0.049$). Of 70 patients with systemic inflammatory response syndrome, 20 had sepsis. Histidine-rich glycoprotein levels were lower in septic patients than in noninfective systemic inflammatory response syndrome patients (8.71 µg/mL [6.72–15.74 µg/mL] vs 33.27 µg/mL [26.57–44.99 µg/mL]; $p < 0.001$) and were lower in nonsurvivors ($n = 8$) than in survivors ($n = 62$) of systemic inflammatory response syndrome (9.06 µg/mL [4.49–15.70 µg/mL] vs 31.78 µg/mL [18.57–42.11 µg/mL]; $p < 0.001$). Histidine-rich glycoprotein showed a high sensitivity and specificity for diagnosing sepsis. Receiver operating characteristic curve analysis for detecting sepsis within systemic inflammatory response syndrome patients showed that the area under the curve for histidine-rich glycoprotein, procalcitonin, and presepsin was 0.97, 0.82, and 0.77, respectively. In addition, survival analysis in systemic inflammatory response syndrome patients revealed that the Harrell *C*-index for histidine-rich glycoprotein, procalcitonin, and presepsin was 0.85, 0.65, and 0.87, respectively.

Conclusions: Histidine-rich glycoprotein levels were low in patients with sepsis and were significantly related to mortality in systemic inflammatory response syndrome population. Furthermore, as a biomarker, histidine-rich glycoprotein may be superior to procalcitonin and presepsin. (*Crit Care Med* 2018; XX:00–00)

Key Words: biomarker; histidine-rich glycoprotein; sepsis; systemic inflammatory response syndrome

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Sepsis is a systemic illness, one of the most severe diseases of patients encountered in the ICU. Despite recent medical progress, the mortality rate of patients with sepsis shows little sign of improvement (1, 2). There are many clinical biomarkers available for rapid diagnosis of sepsis; however, few have become standard (3). Procalcitonin, the soluble triggering receptor expressed on myeloid cells-1, and presepsin are considered promising diagnostic and prognostic biomarkers, although they are limited in their ability to diagnose sepsis (3, 4).

The liver produces histidine-rich glycoprotein (HRG) present in plasma. HRG is a multidomain (structured) protein that interacts with many ligands and is therefore thought to be involved in many functions such as coagulation, immune response, angiogenesis modulation, and others (5, 6). In particular, some articles previously reported that HRG, both in vitro and in vivo, was highly relevant to infections caused by bacteria (7) and fungi (8) and suggested that HRG plays a protective role in the host defense mechanism (7–9). Recently, our group demonstrated that HRG at normal physiologic concentrations maintains circulating neutrophils and vascular endothelial cells quiescent and that plasma HRG levels are decreased rapidly in mice with sepsis, triggering a cascade of events in septic pathogenesis including immunothrombosis, acute respiratory distress syndrome (ARDS), and disseminated intravascular coagulation (DIC) (10). Based on these findings, we suggested a supplementary therapy with HRG for the treatment of sepsis (10).

In this study, we developed a new enzyme-linked immunosorbent assay (ELISA) to measure HRG levels in plasma and used it to perform a prospective observational study of patients with systemic inflammatory response syndrome (SIRS). We aimed to determine whether there was a difference between HRG levels of patients with and without infection as well as survivors and nonsurvivors.

MATERIALS AND METHODS

Study Design

We conducted a single-center, prospective, and observational investigation that was approved by the Institutional Review Board of the Okayama University Graduate School of Medicine, Dentistry, and Pharmaceutical Sciences. We followed guidelines as outlined in Strengthening the Reporting of Observational Studies in Epidemiology (11).

Patients and Data Collection

Patients newly admitted to the ICU of Okayama University Hospital were prospectively enrolled in the study if they fulfilled at least two diagnostic criteria for SIRS. Inclusion criteria were patients who were expected to stay in the ICU for more than 3 days (excluding less severe patients) and those with an arterial blood collection catheter. Exclusion criteria were less than 20 years old, pregnancy, only overnight stay in the ICU, or failure to obtain consent. For comparison, we collected blood samples from patients fulfilled all inclusion criteria except for SIRS criteria (non-SIRS ICU patients). In addition, plasma samples from healthy volunteers were collected and analyzed to determine HRG levels.

Clinical and laboratory data were collected daily while patients were in the ICU. Initial Sequential Organ Failure Assessment (SOFA) and Acute Physiology and Chronic Evaluation (APACHE) II scores were calculated using clinical variables and blood test results. SIRS, sepsis, severe sepsis, and septic shock were classified according to the guidelines of the American College of Chest Physicians/Society of Critical Care Medicine and the International Surviving Sepsis Campaign Guidelines Committee (Sepsis-2) (12, 13). Follow-up investigations were conducted 28 and 90 days after enrollment and at ICU discharge to determine survivors and nonsurvivors.

Analytical Methods

Blood samples were collected in tubes containing K_2EDTA (BD 367840; Beckton-Dickinson, Franklin Lakes, NJ) within 24 hours of ICU admission, processed within 30 minutes of sampling, and centrifuged at 3,500 rpm for 10 minutes. The supernatant was pipetted into polypropylene tubes, a protease inhibitor cocktail (Complete mini EDTA-free; Roche Diagnostics, Basel, Switzerland) was added, and samples were stored at –80°C.

HRG levels were determined using the quantitative sandwich ELISA with a rat monoclonal antibody (mAb) against human HRG (made in-house, number 75-14) as the capture antibody and horseradish peroxidase-conjugated nickel-nitrilotriacetic acid (Ni-NTA HRP Conjugate; Qiagen, Venlo, the Netherlands) for detection. To perform ELISA, 3 µg of mAb per well was diluted in coating buffer (0.05M Na_2Co_3, pH 9.6) and immobilized on a 96-well plate (COSTAR 3590; Corning, Tewksbury, MA) overnight at 4°C. After three washing steps using phosphate-buffered saline (PBS) containing 0.05% Tween 20, the plate was incubated with blocking buffer containing 3% bovine serum albumin (BSA) in PBS for 1 hour at 37°C. After three further washing steps, plasma samples were diluted 1:50 in PBS containing 1% BSA and incubated for 2 hours at 37°C in the mAb-coated wells on the microplate shaker set at 500 rpm. After three washing steps, the plate was incubated with the Ni-NTA HRP conjugate diluted 1:1,000 in PBS containing 0.2% BSA for 1.5 hours at 37°C with shaking. After six extensive washing steps, o-phenylenediamine (Wako, Osaka, Japan) and stop solution (3M H_2So_4) were added, and absorbance at 492 nm was measured using a 96-well plate reader (Model 680; Bio-Rad, Hercules, CA). A standard curve (**Supplementary Fig. 1**, Supplemental Digital Content 1, http://links.lww.com/CCM/D212; **legend**, Supplemental Digital Content 7, http://links.lww.com/CCM/D218) was established using serial dilutions of known amounts of purified HRG (made in-house) (**Supplementary Fig. 2**, Supplemental Digital Content 2, http://links.lww.com/CCM/D213; legend, Supplemental Digital Content 7, http://links.lww.com/CCM/D218). Intraassay reproducibility was determined by assaying the sample six times, and interassay reproducibility was determined by five independent assays. The intraassay and interassay coefficients of variability were 4.19% and 15.5%, respectively. Duplicate plasma samples were tested, and independent assays were repeated twice.

Procalcitonin levels were determined using an automated electrochemiluminescence immunoanalyzer (Modular Analytics E-170; Roche Diagnostics, Mannheim, Germany) in the Clinical Chemistry Laboratory of Okayama University Hospital. Presepsin levels were measured using PATHFAST Presepsin (LSI Medience, Tokyo, Japan).

Outcomes

The primary outcome of this study was to assess the significance of the difference between HRG levels in healthy volunteers, non-SIRS patients, and patients with SIRS. Secondary outcomes were to assess differences between each marker in patients with and without sepsis as well as survivors and nonsurvivors of SIRS.

Statistical Analysis

Data were expressed as median and interquartile ranges (IQRs, 25–75th percentiles), all analyses were two sided, and a p value of less than 0.05 was considered statistically significant. The Mann-Whitney U test or the Kruskal-Wallis test implemented following the Steel-Dwass method was used to compare groups. The receiver operating characteristic (ROC) curve analysis was used to determine the diagnostic accuracy. The Cox proportional hazard model and Kaplan-Meier method were used to analyze survival. We performed survival analysis using the 90-day mortality. In addition, we made adjustment with APACHE II score to correct for disease severity. When we performed Kaplan-Meier method, we divided patients into two groups according to the cutoff value which was calculated in logistic regression model (sensitivity analysis); hazard ratio (HR) was calculated with Cox proportional hazard model. We calculated Spearman rank correlation coefficient to assess correlations between HRG and other variables. We used JMP Pro 11 software (SAS Institute, Chicago, IL) for all analyses, except for calculations of the Harrell C-index, which was determined using STATA 12 software (SAS Institute).

RESULTS

Patient Characteristics

SIRS patients were prospectively included from November 2012 to November 2014. During this period, the ICUs of Okayama University Hospital admitted 3,664 patients, including 728 with SIRS. About three fourth of them were one night stay in ICU. For lack of resource availability, we were granted written consent to collect blood from 70 patients, whose characteristics are shown in **Table 1** and **Supplementary Table 1** (Supplemental Digital Content 3, http://links.lww.com/CCM/D214). The median age of patients was 67 years (IQR, 62–76 yr), 52 (74%) were males, and all were treated in the ICU for 6 days (IQR, 4–9 d). The median patient APACHE II and SOFA scores were 15 (IQR, 12.7–18.2) and 3 (IQR, 2–5), respectively, and 20 patients (29%) were diagnosed with sepsis. ICU mortality and 90-day mortality were both 11% (eight patients), and 28-day mortality was 7.1% (five patients). We were granted written consent from nine non-SIRS patients and 16 healthy volunteers. There were no differences in age among all groups used in analyses, except for healthy volunteers.

Plasma Levels of HRG and Other Markers

The median HRG level in healthy volunteers (n = 16) was 63.00 µg/mL (IQR, 51.53–66.21 µg/mL) (**Supplementary Fig. 3*A***, Supplemental Digital Content 4, http://links.lww.com/CCM/D215; legend, Supplemental Digital Content 7, http://links.lww.com/CCM/D218). HRG levels in non-SIRS patients (n = 9; 38.64 µg/mL [IQR, 30.26–51.81 µg/mL]) were significantly lower than those in healthy volunteers (p = 0.0017). Furthermore, HRG levels in SIRS patients (n = 70; 28.72 µg/mL [IQR, 15.74–41.46 µg/mL]) were lower than those in non-SIRS patients (p = 0.049).

Supplementary Fig. 3, ***B*** and ***C*** (Supplemental Digital Content 4, http://links.lww.com/CCM/D215; legend, Supplemental Digital Content 7, http://links.lww.com/CCM/D218) show the results of secondary analyses. Comparison of patients with sepsis (n = 20) and patients with noninfective SIRS (n = 50) showed that HRG levels in the former group were significantly lower than those in the latter group (8.71 µg/mL [IQR, 6.72–15.74 µg/mL] vs 33.27 µg/mL[IQR, 26.57–44.99 µg/mL]; p < 0.001). Furthermore, procalcitonin and presepsin levels of septic patients were significantly higher than those of noninfective SIRS patients. In SIRS patients, HRG levels of nonsurvivors (n = 8; 9.06 µg/mL [IQR, 4.49–15.70 µg/mL]) were significantly lower (p < 0.001) than those of survivors (n = 62; 31.78 µg/mL [IQR, 18.57–42.11 µg/mL]). Although presepsin levels of nonsurvivors (1,276 pg/mL[IQR, 802.7–5437 pg/mL]) were significantly higher (p < 0.001) than those of survivors (449 pg/mL [IQR, 326.7–618.7 pg/mL]), their procalcitonin levels (0.520 ng/mL [IQR, 0.220–1.277 ng/mL] vs 1.605 ng/mL [IQR, 0.555–3.330 ng/mL]) were not significantly different (p = 0.73). Within septic patients, there were no differences in HRG level between survivors (n = 12) and nonsurvivors (n = 8) (data not shown).

Diagnostic Accuracy of HRG Levels

We performed ROC curve analysis to detect patients with sepsis within the group with SIRS. The ROC curve for HRG was highly sensitive and specific, with the following area under the curve (AUC) values: HRG, 0.97; procalcitonin, 0.82; presepsin, 0.77 (**Fig. 1**). AUC for HRG was higher than that of procalcitonin (p = 0.0018) and presepsin (p = 0.0012).

Associations Between Markers and Mortality

Table 2 shows associations between the plasma level of each marker and mortality. HRG level on ICU day 1 was significantly associated with mortality (HR, 0.88; 95% CI, 0.80–0.98; p < 0.001), and when adjusted according to APACHE II score, this level remained an independent prognostic factor (adjusted HR, 0.89; 95% CI, 0.78–0.97; p = 0.0053). The presepsin level was significantly associated with mortality in univariate analysis (HR, 1.03; 95% CI, 1.01–1.05; p = 0.0040), although when adjusted using the APACHE II score, there was no significant association between presepsin level and mortality (adjusted

TABLE 1. Patient Characteristics

Variables	Healthy Volunteers	Non-SIRS Patients	SIRS Patients		
			Total	Noninfective SIRS	Sepsis
n	16	9	70	50	20
Age, yr, median (IQR)	31.5 (25.7–35.7)	68.0 (66.0–72.5)	67 (62–76)	66.5 (62.7–74.5)	68 (60.5–77)
Male sex, *n* (%)	12 (75)	6 (66.7)	52 (74.2)	39 (78)	13 (65)
ICU death, *n* (%)		0	8 (11)	0	8 (40)
28-d death, *n* (%)		0	5 (7.1)	0	5 (25)
90-d death, *n* (%)		0	8 (11)	0	8 (40)
ICU stay, d, median (IQR)		5 (3.5–6.5)	6 (4–9)	6 (4–7.2)	14 (6.25–26.5)
Severity of disease					
APACHE II score, median (IQR)		13.0 (10.5–15.5)	15 (12.7–18.2)	14 (12–16)	19.5 (17.2–28.5)
SOFA score, median (IQR)		1 (0–2)	3 (2–5)	2 (2–4)	8 (5.2–12)
Severe sepsis, *n* (%)		0	9 (13)	0	9 (45)
Septic shock, *n* (%)		0	8 (11)	0	8 (40)
Medical patients, *n* (%)		0	16 (23)	1 (2)	15 (75)
Pneumonia		0	6	0	6
Renal failure		0	3	0	3
Hepatic failure		0	1	0	1
Pancreatitis		0	1	0	1
Brain infarction		0	1	1	0
Ileus		0	1	0	1
Others		0	3	0	3
Surgical patients, *n* (%)		9	54 (77)	49 (98)	5 (25)
Abdominal		1	8	3	5
Esophageal		4	21	21	0
Laryngeal		3	18	18	0
Hepatic		0	2	2	0
Others		1	5	5	0
Ventilation days, median (IQR)		1 (1)	1 (0–1)	1 (1)	0 (0–7.5)
Vasopressors, *n* (%)		0	10 (14)	2 (4.0)	8 (40)
Blood purification, *n* (%)		0	7 (10)	0	7 (35)
Corticosteroids, *n* (%)		0	9 (13)	1 (2.0)	8 (40)

IQR = interquartile range, SIRS = systemic inflammatory response syndrome.

HR, 0.99; 95% CI, 0.99–1.00; $p = 0.49$). The procalcitonin level did not significantly associate with mortality. The Harrell *C*-index for mortality was as follows: HRG, 0.85; procalcitonin, 0.65; presepsin, 0.87; APACHE II score, 0.90; SOFA score, 0.88; *C*-reactive protein (CRP), 0.61.

Supplementary Table 2 (Supplemental Digital Content 5, http://links.lww.com/CCM/D216) shows that the sensitivity and specificity of HRG levels associated with mortality at the cutoff level of 16.0 μg/mL were 0.87 and 0.79, respectively. Thus, when patients were divided into higher HRG and lower HRG groups according to this cutoff level, Kaplan-Meier curves (**Fig. 2**) showed that the mortality of the lower HRG group was significantly higher than that of the higher HRG group (HR, 9.18; 95% CI, 1.85–45.5; $p = 0.0028$).

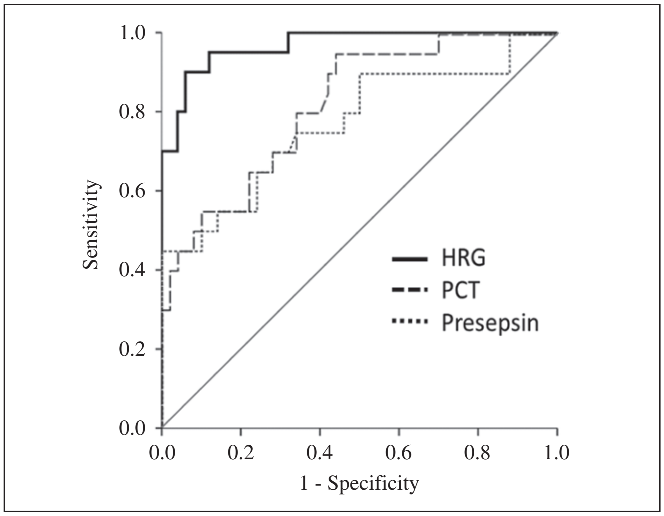

Figure 1. Receiver operating characteristic (ROC) curve analysis for detecting sepsis. ROC curves of histidine-rich glycoprotein (HRG), procalcitonin (PCT), and presepsin. The area under the curve (AUC) in ROC curve analysis for HRG was 0.97. AUC for HRG was higher than that of PCT (0.82; $p = 0.0018$) and presepsin (0.77; $p = 0.0012$).

DISCUSSION

In this study, we found that HRG levels of SIRS patients were significantly lower than those of non-SIRS patients and that HRG levels of septic patients were lower than those of non-infective SIRS patients. In addition, HRG was significantly associated with mortality and provided sufficient diagnostic and prognostic accuracy as a biomarker for sepsis within SIRS patients.

HRG levels decreased in patients with SIRS who were treated in the ICU. To our knowledge, there are no reports describing HRG levels in critically ill patients, although HRG levels have been shown to decrease in patients with liver insufficiency (14) and in those receiving corticosteroids (15). HRG levels have also been proposed to decrease during pregnancy and further decrease in patients with preeclampsia (16). In addition, HRG levels have been shown to decrease in outpatients with elevated CRP values, leading to the conclusion that HRG acts as a negative acute phase reactant (17). In this study, we demonstrated that HRG levels decreased in patients with SIRS and were negatively correlated with CRP levels (**Supplementary Table 3**, Supplemental Digital Content 6, http://links.lww.com/CCM/D217). Our results support the conclusion that inflammation decreases HRG levels.

We evaluated HRG as a biomarker for sepsis by comparing HRG levels with levels of procalcitonin and presepsin, both clinical biomarkers for sepsis (3). When we divided SIRS patients into groups with and without infection, ROC curve analysis for diagnosing sepsis revealed that AUC for HRG, procalcitonin, and presepsin was 0.97, 0.82, and 0.77, respectively. These data indicate that HRG is the best marker for detecting sepsis within SIRS patients. The review article focused on the use of procalcitonin in septic patients in an ICU setting reported that the sensitivity to detect sepsis ranged from 65% to 96% and the specificity ranged from 70% to 89%, which was in agreement with current study (18).

Furthermore, we demonstrated that HRG and presepsin levels, but not procalcitonin levels, were associated with mortality. The Harrell *C*-index (predictive power) for mortality was 0.85 and 0.87 for HRG and presepsin, respectively, consistent with the *C*-index for the APACHE II score (0.90). This score is an established clinical prognostic marker used worldwide but involves a complicated scoring system calculated according to dozens of variables. Thus, our present results strongly suggest that HRG will serve as a more effective prognostic biomarker for SIRS patients.

Using a mouse sepsis model, we clearly demonstrated that plasma HRG decreased markedly due to reduction of messenger RNA expression in the liver, degradation by thrombin, and deposition on intravascular thrombi (10). Under such condition, a cascade of responses including intravascular neutrophil extracellular traps formation, strong attachment of neutrophils to vascular endothelial cells, and immunothrombus formation proceed, leading to ARDS, DIC, and multiple organ failure (10). Thus, the marked decrease in plasma HRG may have a direct causal relation with septic lethality.

In this study, we also developed sandwich ELISA to measure HRG levels using one mAb to capture HRG, taking advantage of the high-affinity binding of HRG to Ni–NTA (19). This method does not rely on different HRG epitopes and therefore does not require the use of multiple antibodies. Using this ELISA, we determined that the median HRG level in healthy volunteers was 63.00 µg/mL (IQR, 51.53–66.21 µg/mL), in agreement with published data showing that HRG levels are approximately 100 µg/mL in human plasma and that they vary widely (5, 6). We therefore conclude that HRG ELISA developed in this study is acceptable for clinical practice.

TABLE 2. Associations Between Each Marker and Mortality

Variables	Univariate Analysis			Adjusted With Acute Physiology and Chronic Evaluation II Score	
	HR (95% CI)	*p*	Harrell *C*-index	Adjusted HR (95% CI)	*p*
Histidine-rich glycoprotein	0.88 (0.80–0.98)	< 0.001	0.85	0.89 (0.78–0.97)	0.0053
Procalcitonin	0.97 (0.84–1.13)	0.76	0.65		
Presepsin	1.03 (1.01–1.05)	0.0040	0.87	0.99 (0.99–1.00)	0.49

HR = hazard ratio.
Adjusted HR denotes HR adjusted according to Acute Physiology and Chronic Evaluation II score.

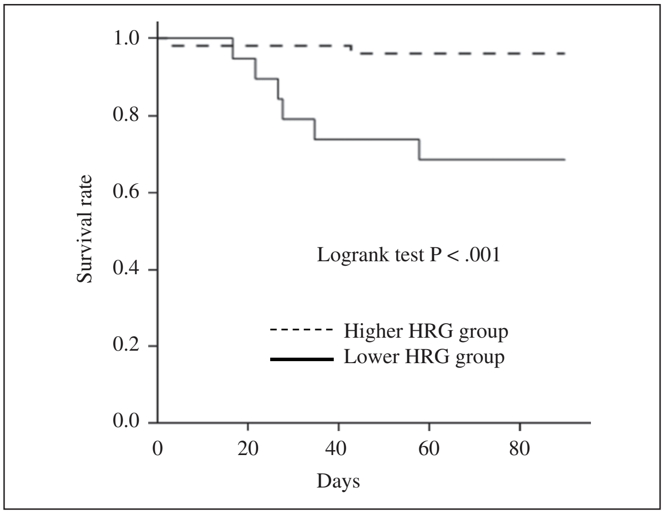

Figure 2. Kaplan-Meier survival curves. Patients were divided into higher histidine-rich glycoprotein (HRG) and lower HRG groups according to the cutoff level of 16.0 µg/mL. At the cutoff level of 16.0 µg/mL, the sensitivity and specificity of HRG levels associated with mortality were 0.87 and 0.79, respectively.

There are limitations to this study. First, our study was a single-center study and included only 79 patients. However, to address this limitation, we initiated a multicenter prospective study to validate the diagnostic and prognostic role of HRG levels. Second, although we focused on sepsis, we studied 70 patients with SIRS, including only 20 who had the disease. Thus, it is unclear whether HRG would work to differentiate sepsis survival due to limited numbers of individual analyzed here. Larger validation studies focused on sepsis should be performed. Third, in this study, we used old definition of sepsis (Sepsis-2) because we conducted this study from November 2012 to November 2014. We should initiate another study with new definition for sepsis (Sepsis-3) (20). Fourth, we only assessed the initial HRG level, and we had no serial data. Time-dependent changes in HRG levels would be more valuable and reflect treatments such as steroids and renal replacement therapies. Further investigations about serial values would be needed. Fifth, noninfective SIRS patients included many postoperative patients. Postoperative condition is complicated because of the effects of general anesthesia, surgical pain, or something associated with operation. These effects might confuse a comparison between noninfective SIRS patients and septic patients. Sixth, noninfective SIRS patients included many patients with cancer. Because HRG levels may prevent tumor growth (5, 21, 22) and are significantly higher in patients with breast cancer (23), the characteristics of these populations may have influenced our data.

CONCLUSIONS

HRG levels of septic patients were significantly lower than those of noninfective SIRS patients, and HRG levels were significantly associated with mortality within the SIRS population. Therefore, HRG may be superior to procalcitonin and presepsin for assessing severity of SIRS patients. Our results suggest that HRG serves as a novel biomarker for diagnosing sepsis, evaluating severity of patients, and predicting patient outcomes. To confirm our findings, larger validation studies are needed.

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REFERENCES

1. Russell JA: Management of sepsis. *N Engl J Med* 2006; 355:1699–1713
2. Lever A, Mackenzie I: Sepsis: Definition, epidemiology, and diagnosis. *BMJ* 2007; 335:879–883
3. Henriquez-Camacho C, Losa J: Biomarkers for sepsis. *Biomed Res Int* 2014; 2014:547818
4. Pierrakos C, Vincent JL: Sepsis biomarkers: A review. *Crit Care* 2010; 14:R15
5. Poon IK, Patel KK, Davis DS, et al: Histidine-rich glycoprotein: The Swiss Army knife of mammalian plasma. *Blood* 2011; 117:2093–2101
6. Blank M, Shoenfeld Y: Histidine-rich glycoprotein modulation of immune/autoimmune, vascular, and coagulation systems. *Clin Rev Allergy Immunol* 2008; 34:307–312
7. Shannon O, Rydengård V, Schmidtchen A, et al: Histidine-rich glycoprotein promotes bacterial entrapment in clots and decreases mortality in a mouse model of sepsis. *Blood* 2010; 116:2365–2372
8. Rydengård V, Shannon O, Lundqvist K, et al: Histidine-rich glycoprotein protects from systemic Candida infection. *PLoS Pathog* 2008; 4:e1000116
9. Priebatsch KM, Kvansakul M, Poon IK et al: Functional regulation of the plasma protein histidine-rich glycoprotein by Zn2+ in settings of tissue injury. *Biomolecules* 2017; 7:E22
10. Wake H, Mori S, Liu K, et al: Histidine-rich glycoprotein prevents septic lethality through regulation of immunothrombosis and inflammation. *EBioMedicine* 2016; 9:180–194
11. Centre C, Vandenbroucke JP: Strengthening the reporting of observational studies in epidemiology (STROBE) statement: Guidelines for reporting observational studies. *BMJ* 2007; 335:806–808
12. Levy MM, Fink MP, Marshall JC, et al: 2001 SCCM/ESICM/ACCP/ATS/SIS international sepsis definitions conference. *Crit Care Med* 2003; 31:1250–1256
13. Dellinger RP, Levy MM, Rhodes A, et al; Surviving Sepsis Campaign Guidelines Committee including the Pediatric Subgroup: Surviving sepsis campaign: International guidelines for management of severe sepsis and septic shock: 2012. *Crit Care Med* 2013; 41:580–637
14. Lijnen HR, Jacobs G, Collen D: Histidine-rich glycoprotein in a normal and a clinical population. *Thromb Res* 1981; 22:519–523
15. Morgan WT: Serum histidine-rich glycoprotein levels are decreased in acquired immune deficiency syndrome and by steroid therapy. *Biochem Med Metab Biol* 1986; 36:210–213
16. Bolin M, Akerud P, Hansson A, et al: Histidine-rich glycoprotein as an early biomarker of preeclampsia. *Am J Hypertens* 2011; 24:496–501

17. Saigo K, Yoshida A, Ryo R, et al: Histidine-rich glycoprotein as a negative acute phase reactant. *Am J Hematol* 1990; 34:149–150
18. Carr JA: Procalcitonin-guided antibiotic therapy for septic patients in the surgical intensive care unit. *J Intensive Care* 2015; 3:36
19. Mori S, Takahashi HK, Yamaoka K, et al: High affinity binding of serum histidine-rich glycoprotein to nickel-nitrilotriacetic acid: The application to microquantification. *Life Sci* 2003; 73:93–102
20. Singer M, Deutschman CS, Seymour CW, et al: The third international consensus definitions for sepsis and septic shock (Sepsis-3). *JAMA* 2016; 315:801–810
21. Johnson LD, Goubran HA, Kotb RR: Histidine rich glycoprotein and cancer: A multi-faceted relationship. *Anticancer Res* 2014; 34:593–603
22. Cedervall J, Zhang Y, Ringvall M, et al: HRG regulates tumor progression, epithelial to mesenchymal transition and metastasis via platelet-induced signaling in the pre-tumorigenic microenvironment. *Angiogenesis* 2013; 16:889–902
23. Matboli M, Eissa S, Said H: Evaluation of histidine-rich glycoprotein tissue RNA and serum protein as novel markers for breast cancer. *Med Oncol* 2014; 31:897